Lecture 2

A driven qubit: magnetic resonance

Objectives. *A proton in a strong magnetic field is a qubit with two energy levels. A second field, two thousand times weaker, can turn its spin over completely, provided that it rotates at the right frequency. This lecture explains how. The weak field changes in time, and so does the Hamiltonian. We solve the Schrödinger equation exactly in a frame that turns with the field, and find Rabi's formula for the probability that the spin has turned over. Two short pulses with a long wait between them then give fringes far narrower than the resonance.*

By the end of this lecture you will be able to:

- Solve the Schrödinger equation of a circular drive in the rotating frame.
- Find, from Rabi's formula, the probability that a spin turns over, on and off resonance, and the width of the resonance.
- Choose the duration and phase of a resonant pulse that takes a spin prepared up to a given state of the rotating frame.
- Find the probability that the spin turns over after Ramsey's two pulses, and the width of the central fringe.

2.1 Turning a spin over

We write $|\uparrow\rangle = |0\rangle$ and $|\downarrow\rangle = |1\rangle$ for the states whose Bloch vectors are $\mathbf{e}_z$ and $-\mathbf{e}_z$, spin up and spin down along z. By Lecture 1, a spin in a strong magnetic field along the z-axis has the Hamiltonian $H = \frac{\hbar\nu_0}{2}\sigma_z$ with $\nu_0 > 0$. The Bloch vector precesses about $\mathbf{e}_z$ at the angular frequency ν_0. The states $|\uparrow\rangle$ and $|\downarrow\rangle$ are the stationary states of H. How can we take a spin from $|\uparrow\rangle$ to $|\downarrow\rangle$?

By Theorem 1.12, a constant Hamiltonian keeps the component $\mathbf{n}\cdot\mathbf{r}$ constant. It can turn $\mathbf{e}_z$ into $-\mathbf{e}_z$ only if $\mathbf{n}\cdot\mathbf{e}_z = -\mathbf{n}\cdot\mathbf{e}_z$, that is, only if its axis $\mathbf{n}$ is perpendicular to $\mathbf{e}_z$. Add a weak static field along x. The Hamiltonian becomes

$$H = \frac{\hbar}{2}\big(\nu_0\sigma_z + \Omega\sigma_x\big), \qquad 0 < \Omega \ll \nu_0.$$

Its axis is the unit vector $\mathbf{n}$ in the direction of $(\Omega, 0, \nu_0)$, tilted from $\mathbf{e}_z$ by the small angle α with $\tan\alpha = \Omega/\nu_0$. The precession keeps the component $\mathbf{n}\cdot\mathbf{r}$ constant. So the Bloch vector of a spin prepared up stays on the circle of latitude about $\mathbf{n}$ through $\mathbf{e}_z$. This circle consists of the points $\mathbf{r}$ of the sphere with $\mathbf{n}\cdot\mathbf{r} = \mathbf{n}\cdot\mathbf{e}_z$ (Remark 1.9). The lowest point of that circle has $r_z = \cos 2\alpha$. By Corollary 1.8 with $\mathbf{m} = \mathbf{e}_z$, the probability of reading $|\downarrow\rangle$ is $\frac{1}{2}(1 - r_z)$, so it never exceeds

$$\frac{1}{2}(1 - \cos 2\alpha) = \sin^2\alpha = \frac{\Omega^2}{\Omega^2 + \nu_0^2}.$$

For $\Omega/\nu_0 = 5 \cdot 10^{-4}$ this bound is $2.5 \cdot 10^{-7}$. A weak static field tilts the axis a little, so the spin barely moves.

A field that turns does better. The spin precesses about $\mathbf{e}_z$ at the rate ν_0. A transverse field that turns about $\mathbf{e}_z$ at the same rate keeps a fixed angle to the precessing spin, so its effect builds up instead of averaging out. The rest of the lecture computes how far it builds up.

2.2 A Hamiltonian that changes in time

The coils of the experiment make a transverse field of fixed strength that turns about $\mathbf{e}_z$ at the angular frequency $\nu > 0$. The Hamiltonian is

$$H(t) = \frac{\hbar\nu_0}{2}\,\sigma_z + \frac{\hbar\Omega}{2}\big(\cos(\nu t)\,\sigma_x + \sin(\nu t)\,\sigma_y\big), \qquad \nu_0 > 0,\ \Omega > 0. \tag{2.1}$$

A field that the experimenter imposes and changes in time is called a *drive*. This one is a *circular drive*, since the field turns on a circle. The number Ω is the *Rabi frequency*, the precession rate that the transverse field alone would cause. The field is set by the experimenter, and we treat it as a given function of time, not as a quantum system.

Postulate 1.10 holds with $H(t)$ in place of H:

$$\mathrm{i}\hbar\,\frac{\mathrm{d}\psi_t}{\mathrm{d}t} = H(t)\psi_t.$$

At every time the Hamiltonian $H(t)$ is self-adjoint. The proof of Lemma 1.11 differentiates $\langle\psi_t, \psi_t\rangle$ at one time and uses only that the Hamiltonian is self-adjoint at that time. So the norm $\|\psi_t\|$ is constant here too, and the total probability stays 1.

Lemma 1.11 also solved the equation, by the exponential of the constant matrix $-\mathrm{i}tH/\hbar$ (*Linear Algebra*, Prop. 6.21). That proposition needs a constant matrix, and $H(t)$ is not constant. We shall not need a theory of equations with variable coefficients. A change of variables removes the time dependence of the Hamiltonian (2.1) exactly.

2.3 The rotating frame

Let

$$R(t) = e^{-\mathrm{i}\nu t\sigma_z/2} = \begin{pmatrix} e^{-\mathrm{i}\nu t/2} & 0 \\ 0 & e^{\mathrm{i}\nu t/2} \end{pmatrix}.$$

By Lemma 1.11, the matrix $R(t)$ is the evolution under the Hamiltonian $\frac{\hbar\nu}{2}\sigma_z$ for the time t. So by Theorem 1.12 it turns the Bloch sphere about $\mathbf{e}_z$ through the angle νt:

$$\mathbf{r}\big(R(t)\chi\big) = \mathrm{Rot}_z(\nu t)\,\mathbf{r}(\chi) \qquad \text{for every unit vector } \chi, \tag{2.2}$$

where $\mathrm{Rot}_z(\beta)$ is the rotation of $\mathbb{R}^3$ about $\mathbf{e}_z$ through the angle β, anticlockwise seen from the tip of $\mathbf{e}_z$. Put $\Delta = \nu_0 - \nu$, the *detuning* of the drive, and

$$\tilde{H} = \frac{\hbar}{2}\big(\Delta\,\sigma_z + \Omega\,\sigma_x\big). \tag{2.3}$$

Lemma 2.1 (The rotating frame). *Let $H(t)$ be the Hamiltonian (2.1).*

1. $H(t) = R(t)\,\frac{\hbar}{2}\big(\nu_0\sigma_z + \Omega\sigma_x\big)\,R(t)^\dagger$.
2. *A differentiable ψ_t solves $\mathrm{i}\hbar\,\dot{\psi}_t = H(t)\psi_t$ if and only if the function $\tilde{\psi}_t = R(t)^\dagger\psi_t$ solves*

$\mathrm{i}\hbar\,\dot{\tilde\psi}_t = \tilde H\tilde\psi_t$.

3. *For every unit vector ψ, the only solution ψ_t with $\psi_0 = \psi$ is*

$$\psi_t = R(t)\, e^{-\mathrm{i}t\tilde H/\hbar}\,\psi. \tag{2.4}$$

Proof.

1. The matrix $R = R(t)$ is diagonal, so $R\sigma_z R^\dagger = \sigma_z$. Next,

$$R\,\sigma_x R^\dagger = \begin{pmatrix} 0 & e^{-\mathrm{i}\nu t} \\ e^{\mathrm{i}\nu t} & 0 \end{pmatrix} = \cos(\nu t)\,\sigma_x + \sin(\nu t)\,\sigma_y.$$

   Multiplying by $\frac{\hbar\Omega}{2}$ and adding $\frac{\hbar\nu_0}{2}\sigma_z$ gives the Hamiltonian (2.1).

2. Write $\psi_t = R\tilde\psi_t$. By *Linear Algebra*, Prop. 6.21, the derivative of R is $\dot R = -\frac{\mathrm{i}\nu}{2}\sigma_z R$, so $\mathrm{i}\hbar\dot R = \frac{\hbar\nu}{2}\sigma_z R$. The product rule gives

$$\mathrm{i}\hbar\,\dot\psi_t - H(t)\psi_t = R\Big(\mathrm{i}\hbar\,\dot{\tilde\psi}_t + \tfrac{\hbar\nu}{2}\sigma_z\tilde\psi_t - R^\dagger H(t) R\,\tilde\psi_t\Big),$$

   where we used $\sigma_z R = R\sigma_z$. By the first part, $R^\dagger H(t) R = \frac{\hbar}{2}(\nu_0\sigma_z + \Omega\sigma_x)$. Subtracting $\frac{\hbar\nu}{2}\sigma_z$ from this matrix leaves $\tilde H$. So the bracket is $\mathrm{i}\hbar\dot{\tilde\psi}_t - \tilde H\tilde\psi_t$. The matrix R is invertible, so one side of the identity vanishes exactly when the other does.

3. Since $R(0) = \mathbb{1}$, the two functions start at the same vector ψ. By Lemma 1.11, the equation with the constant Hamiltonian $\tilde H$ has exactly one solution, $\tilde\psi_t = e^{-\mathrm{i}t\tilde H/\hbar}\psi$. By the second part, the solutions of the two equations correspond one to one, so the driven equation also has exactly one, $R(t)\tilde\psi_t$. ∎

The vector $\tilde\psi_t$ describes the state in coordinates that turn with the transverse field. Its Bloch vector $\tilde{\mathbf{r}}$ is related to the Bloch vector of ψ_t by (2.2): $\mathbf{r} = \mathrm{Rot}_z(\nu t)\,\tilde{\mathbf{r}}$. In these coordinates the transverse field stands still along $\mathbf{e}_x$. The term in σ_z has the coefficient Δ in place of ν_0. Without a drive ($\Omega = 0$) the Bloch vector precesses at the rate ν_0, and the vector $\tilde{\mathbf{r}}$ at the rate $\Delta = \nu_0 - \nu$. The rotating frame subtracts its own rate from the rate of precession. A state at rest in the laboratory turns backwards, at the rate $-\nu$, in the frame.

The matrix $R(t)$ is diagonal, so $|\langle\downarrow|\psi_t\rangle| = \big|\langle\downarrow|\tilde\psi_t\rangle\big|$. The same holds for $|\uparrow\rangle$. So the probabilities of a measurement along z are the same in both descriptions.

Starting at a time s instead of 0, the same argument gives $\psi_t = U(t,s)\psi_s$ with

$$U(t,s) = R(t)\, e^{-\mathrm{i}(t-s)\tilde H/\hbar}\, R(s)^\dagger,$$

the *evolution operator* from s to t. It is unitary, as a product of unitary matrices.

2.4 Rabi's formula

Theorem 2.2 (Rabi's formula). *Let $H(t)$ be the Hamiltonian (2.1), and put $\tilde\Omega = \sqrt{\Omega^2 + \Delta^2}$. For a spin prepared in the state $|\uparrow\rangle$ at $t = 0$, a measurement along z at the time t reads $|\downarrow\rangle$ with probability*

$$\mathrm{Prob}(\downarrow, t) = \frac{\Omega^2}{\tilde\Omega^2}\,\sin^2\Big(\frac{\tilde\Omega t}{2}\Big). \tag{2.5}$$

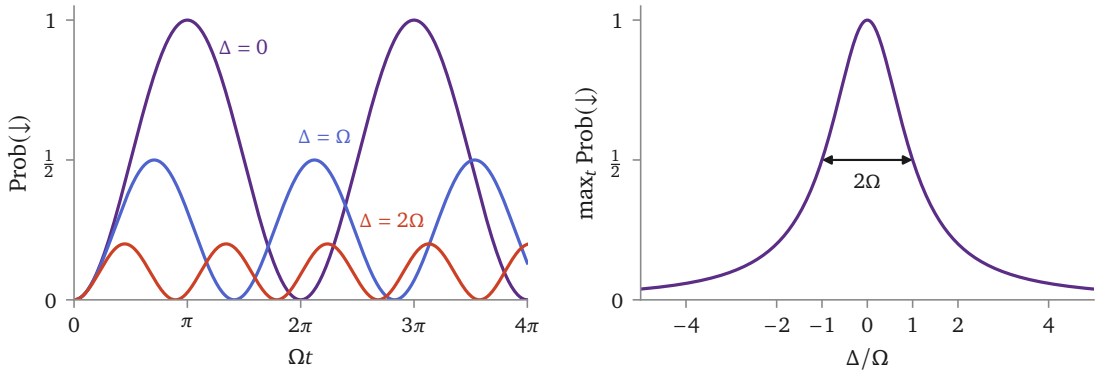


Figure 2.1: Rabi's formula. Left: $\mathrm{Prob}(\downarrow, t)$ against Ωt for $\Delta = 0$, Ω and 2Ω. Right: the largest value $\Omega^2/(\Omega^2 + \Delta^2)$ against Δ/Ω. It is $\frac{1}{2}$ at $\Delta = \pm\Omega$: the full width at half maximum is 2Ω.

Proof. Since $\Omega > 0$, also $\tilde{\Omega} > 0$. So (2.3) reads $\tilde{H} = \frac{\hbar\tilde{\Omega}}{2}\,\tilde{\mathbf{n}} \cdot \boldsymbol{\sigma}$ with the unit vector $\tilde{\mathbf{n}} = (\Omega, 0, \Delta)/\tilde{\Omega}$. By Lemma 2.1 and Theorem 1.12, the Bloch vector $\tilde{\mathbf{r}}$ turns about $\tilde{\mathbf{n}}$ at the rate $\tilde{\Omega}$, starting at $\mathbf{e}_z$. The formula of Theorem 1.12 with $\mathbf{r}_0 = \mathbf{e}_z$ has z-component

$$\tilde{r}_z(t) = \cos(\tilde{\Omega}t) + \big(1 - \cos(\tilde{\Omega}t)\big)\,\frac{\Delta^2}{\tilde{\Omega}^2},$$

since $\tilde{\mathbf{n}} \times \mathbf{e}_z$ has no z-component. The measurement along z has the same statistics in both descriptions, so by Corollary 1.8, using $1 - \Delta^2/\tilde{\Omega}^2 = \Omega^2/\tilde{\Omega}^2$,

$$\mathrm{Prob}(\downarrow, t) = \frac{1}{2}\big(1 - \tilde{r}_z(t)\big) = \frac{\Omega^2}{\tilde{\Omega}^2} \cdot \frac{1 - \cos(\tilde{\Omega}t)}{2} = \frac{\Omega^2}{\tilde{\Omega}^2}\,\sin^2\Big(\frac{\tilde{\Omega}t}{2}\Big).$$

■

Figure 2.1 draws the formula. The drive is *on resonance* when $\nu = \nu_0$, that is, $\Delta = 0$. Then $\tilde{\Omega} = \Omega$, and

$$\mathrm{Prob}(\downarrow, t) = \sin^2\Big(\frac{\Omega t}{2}\Big),$$

which reaches 1 at $t = \pi/\Omega$, however small $\Omega > 0$ is. In the rotating frame the static field is gone, and only the weak field is left. The vector $\tilde{\mathbf{r}}$ turns about $\mathbf{e}_x$, slowly, but all the way to $-\mathbf{e}_z$. This is why a weak field can turn a spin over.

Off resonance the probability never reaches 1. Its largest value, $\Omega^2/(\Omega^2 + \Delta^2)$, is $\frac{1}{2}$ when $|\Delta| = \Omega$. For a weak drive, $\Omega < \nu_0$, this happens at the two drive frequencies $\nu = \nu_0 \pm \Omega$, so the resonance has full width at half maximum 2Ω. The weaker the drive, the sharper the resonance, and the slower the turn. A static field is the case $\nu = 0$, with $\Delta = \nu_0$. Then the formula gives the bound of § 2.1.

A resonant drive switched on for the time π/Ω is a *π-pulse*. It takes $|\uparrow\rangle$ to $|\downarrow\rangle$. In the frame, $e^{-\mathrm{i}\pi\sigma_x/2} = -\mathrm{i}\sigma_x$ and $-\mathrm{i}\sigma_x\,|\uparrow\rangle = -\mathrm{i}\,|\downarrow\rangle$. The diagonal matrix $R(t)$ then multiplies $|\downarrow\rangle$ by a phase only. Figure 2.2 draws the path of the Bloch vector in both descriptions. A drive for half that time, $\pi/(2\Omega)$, is a *$\pi/2$-pulse*. It turns $\tilde{\mathbf{r}}$ about $\mathbf{e}_x$ through a quarter turn, from $\mathbf{e}_z$ to $-\mathbf{e}_y$.

Two limits of the formula should be kept in mind. First, it is for a spin prepared up. Starting from any other state, the vector $\tilde{\mathbf{r}}$ still turns about $\tilde{\mathbf{n}}$ at the rate $\tilde{\Omega}$ (Theorem 1.12), but the probability is a different function of time. Second, a resonant drive with a fixed phase turns $\tilde{\mathbf{r}}$ about $\mathbf{e}_x$ only. Starting from $\mathbf{e}_z$, the vector stays in the plane of $\mathbf{e}_y$ and $\mathbf{e}_z$. So one such pulse does not reach every state.

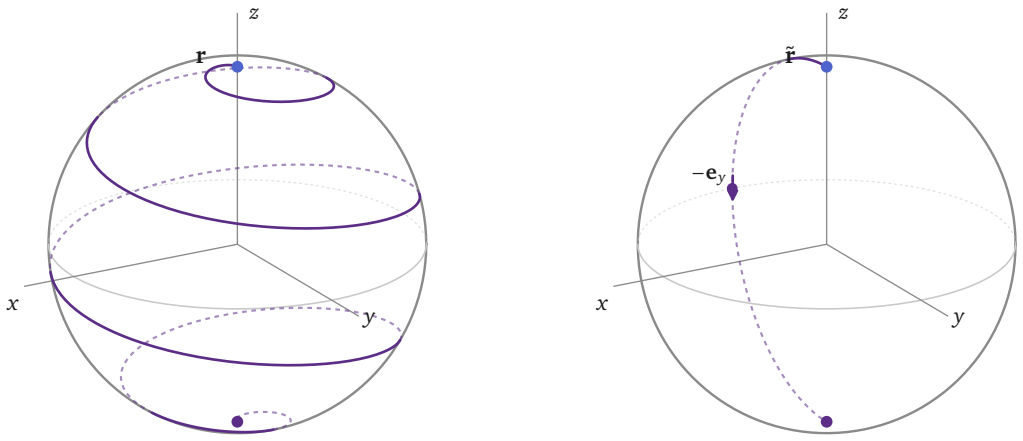


Figure 2.2: One π-pulse on resonance, drawn with $\nu = 8\Omega$. Left: the Bloch vector $\mathbf{r}$ in the laboratory spirals from the north pole to the south pole. Right: in the rotating frame, $\tilde{\mathbf{r}}$ turns about $\mathbf{e}_x$ along half a meridian, through $-\mathbf{e}_y$.

The phase of the drive removes the second limit. Suppose that the coils start the transverse field at the angle ξ, so that $\cos(\nu t)$ and $\sin(\nu t)$ in (2.1) become $\cos(\nu t + \xi)$ and $\sin(\nu t + \xi)$. The proof of Lemma 2.1 goes through with $\cos\xi\,\sigma_x + \sin\xi\,\sigma_y$ in place of σ_x. On resonance, $\tilde{\mathbf{r}}$ then turns about the axis $(\cos\xi, \sin\xi, 0)$. A pulse that starts at $t = 0$ and lasts the time θ/Ω, with $0 \le \theta \le \pi$, takes the vector from $\mathbf{e}_z$ to

$$\tilde{\mathbf{r}} = \big(\sin\theta\sin\xi,\ -\sin\theta\cos\xi,\ \cos\theta\big),$$

the point with polar angle θ and azimuth $\xi - \pi/2$. Every point of the sphere has this form, so a pulse of the right duration and phase takes $\tilde{\mathbf{r}}$ from $\mathbf{e}_z$ to any point of the sphere.

2.5 A proton in an NMR magnet

Example 2.3 (A proton in an NMR magnet).

1. *The Hamiltonian.* A proton has a positive gyromagnetic ratio, $\gamma/2\pi = 42.577$ MHz T^{-1}. By Lecture 1, the spin of a proton in a field $\mathbf{B}$ has the Hamiltonian

   $$H = -\gamma\,\frac{\hbar}{2}\,\boldsymbol{\sigma}\cdot\mathbf{B}.$$

   Take the field

   $$\mathbf{B}(t) = -B_0\,\mathbf{e}_z - B_1\big(\cos(\nu t)\,\mathbf{e}_x + \sin(\nu t)\,\mathbf{e}_y\big), \qquad B_0 = 1\text{ T}, \quad B_1 = 0.5\text{ mT}.$$

   Then $H(t)$ is (2.1) with $\nu_0 = \gamma B_0$ and $\Omega = \gamma B_1$. The static field points along $-\mathbf{e}_z$, because for $\gamma > 0$ the Hamiltonian's axis is $-\mathbf{B}/|\mathbf{B}|$.
2. *The frequencies.* The symbol ν is an angular frequency, so a value in hertz is $\nu/2\pi$. Here $\nu_0/2\pi = 42.58$ MHz and $\Omega/2\pi = 21.29$ kHz, so $\Omega/\nu_0 = B_1/B_0 = 5\cdot 10^{-4}$.
3. *A static field.* A field of the same 0.5 mT, held still along $\mathbf{e}_x$, turns the spin over with probability at most $2.5\cdot 10^{-7}$ (§ 2.1).
4. *On resonance.* A π-pulse lasts $\pi/\Omega = 23.5\ \mu$s, and a $\pi/2$-pulse 11.7 μs.

5. *Off resonance.* A detuning $\Delta = \Omega$ moves $\nu/2\pi$ by 21.3 kHz, five parts in 10^4 of $\nu_0/2\pi$. Then $\tilde{\Omega} = \sqrt{2}\,\Omega$, and the probability of $|\downarrow\rangle$ reaches only $\frac{1}{2}$. It first does so at $t = \pi/(\sqrt{2}\,\Omega) = 16.6\ \mu$s. At $\Delta = 10\,\Omega$ the probability never exceeds $\frac{1}{101} \approx 0.0099$.
6. *After a $\pi/2$-pulse.* On resonance, with the drive switched off at the end of the pulse, the vector $\tilde{\mathbf{r}} = -\mathbf{e}_y$ does not move, since $\tilde{H} = 0$ when $\Delta = \Omega = 0$. By (2.2) the Bloch vector in the laboratory is

$$\mathbf{r}(t) = \mathrm{Rot}_z(\nu t)(-\mathbf{e}_y) = \big(\sin(\nu t),\ -\cos(\nu t),\ 0\big).$$

It turns in the xy-plane at the rate $\nu = \nu_0$.

The physics

Rabi, Zacharias, Millman and Kusch (1938) sent a beam of molecules through a strong magnetic field and a weak oscillating one of fixed radio frequency. When the strong field made the nuclear spins precess at that frequency, the molecules whose spins had turned over were lost from the detected beam. The known radio frequency and the field at the dip in the signal gave the nuclear magnetic moment. In papers published in 1946, Purcell, Torrey and Pound, and independently Bloch, Hansen and Packard, reported the same resonance in bulk matter: paraffin and water. In a water sample at room temperature that has reached thermal equilibrium in the static field, the protons have a small net magnetic moment along the field (Lecture 3). A resonant $\pi/2$-pulse turns this moment into the xy-plane. After the drive is switched off, the moment turns in that plane and induces, in a coil around the sample, a voltage of frequency $\nu_0/2\pi$. That voltage is the signal of nuclear magnetic resonance (NMR), and of the magnetic resonance imaging (MRI) of protons in water.

2.6 Ramsey's two pulses

Rabi's formula locates the frequency ν_0 to within the width 2Ω of the resonance. A sharper resonance needs a weaker drive, and so a longer pulse. In Ramsey's method, two short pulses turn the spin. Between them the drive is switched off, and the spin precesses freely for a long time T. The sharpness of the resonance then comes from T, not from Ω.

The oscillator that feeds the coils keeps running. Only its connection to the coils is switched on and off. So during both pulses the field is the same function (2.1) of the same time t. One rotating frame $R(t)$ serves throughout. The proof of Lemma 2.1 does not use $\Omega > 0$. So while the drive is off, the frame Hamiltonian is $\tilde{H} = \frac{\hbar\Delta}{2}\sigma_z$.

Proposition 2.4 (Ramsey's fringes). *A spin prepared in the state $|\uparrow\rangle$ receives two $\pi/2$-pulses separated by the time T. Between the pulses the drive is off, but its oscillator keeps running, so the drive keeps its phase. Suppose that each pulse turns $\tilde{\mathbf{r}}$ about $\mathbf{e}_x$ through exactly $\pi/2$. Then a measurement along z at the end reads $|\downarrow\rangle$ with probability*

$$\mathrm{Prob}(\downarrow) = \cos^2\Big(\frac{\Delta T}{2}\Big). \tag{2.6}$$

Proof. We follow $\tilde{\mathbf{r}}$ through the three steps.

1. The first pulse takes $\mathbf{e}_z$ to $-\mathbf{e}_y$, as in § 2.4.
2. During the wait, apply Theorem 1.12 with the axis $\mathbf{e}_z$ if $\Delta > 0$ and $-\mathbf{e}_z$ if $\Delta < 0$. If $\Delta = 0$, then $\tilde{H} = 0$ and $\tilde{\mathbf{r}}$ does not move. In every case $\tilde{\mathbf{r}}$ turns about $\mathbf{e}_z$ through the angle ΔT, to $\mathrm{Rot}_z(\Delta T)(-\mathbf{e}_y) = (\sin\Delta T,\ -\cos\Delta T,\ 0)$.

3. A quarter turn about $\mathbf{e}_x$ takes $(x, y, 0)$ to $(x, 0, y)$. So after the second pulse, $\tilde{\mathbf{r}} = (\sin \Delta T,\ 0,\ -\cos \Delta T)$.

The measurement along z has the same statistics in both descriptions (§ 2.3). By Corollary 1.8,

$$\text{Prob}(\downarrow) = \tfrac{1}{2}\big(1 - \tilde{r}_z\big) = \tfrac{1}{2}\big(1 + \cos(\Delta T)\big) = \cos^2\Big(\frac{\Delta T}{2}\Big). \qquad \blacksquare$$

The ideal pulses are an approximation, good when $|\Delta| \ll \Omega$. Then the axis of $\frac{\hbar}{2}(\Delta\sigma_z + \Omega\sigma_x)$ is close to $\mathbf{e}_x$, and each pulse is close to a quarter turn about that axis.

As the drive frequency varies, the probability (2.6) rises and falls in a row of *fringes*. It is 1 when ΔT is a multiple of 2π, and 0 when ΔT is an odd multiple of π. The central fringe falls to half its height at $|\Delta| = \pi/(2T)$, so its full width at half maximum is π/T in Δ. The width is set by the wait T alone, and not by the strength of the drive. All the fringes have the same height, so (2.6) alone does not say which one is the central fringe.

Why does the wait matter, if nothing is driven during it? Throughout the wait $\tilde{r}_z = 0$. A reading along z would then give ± 1 with probability $\frac{1}{2}$, whatever the detuning Δ is. The detuning shows only in the azimuth of $\tilde{\mathbf{r}}$, the relative phase of $|\uparrow\rangle$ and $|\downarrow\rangle$ (Example 1.13). The second pulse turns the angle gained during the wait into a height above the equator, which a reading along z detects.

The physics

Ramsey proposed the method in 1949 for beams of molecules. The two pulses were two short regions of oscillating field at the two ends of the apparatus, and the wait was the molecules' flight between them. Molecules of different speeds spend different times T in flight, so their fringes have different spacings. Averaging over the speeds damps the side fringes, while the central one, at $\Delta = 0$, keeps its full height for every flight time.

2.7 Circular and linear drives

Remark 2.5 (A linear drive). A single coil along x makes a field that oscillates along a line. For the proton of Example 2.3 take the field $-2B_1 \cos(\nu t)\, \mathbf{e}_x$. It adds $\hbar\Omega \cos(\nu t)\, \sigma_x$ to the Hamiltonian, the sum of two circular terms that turn in opposite senses:

$$2\cos(\nu t)\, \sigma_x = \big(\cos(\nu t)\, \sigma_x + \sin(\nu t)\, \sigma_y\big) + \big(\cos(\nu t)\, \sigma_x - \sin(\nu t)\, \sigma_y\big).$$

The rotating frame removes the time dependence of the first one only. In the frame, the second becomes $\frac{\hbar\Omega}{2}\big(\cos(2\nu t)\, \sigma_x - \sin(2\nu t)\, \sigma_y\big)$ and turns at the rate 2ν. Lemma 2.1 and Rabi's formula are exact for a circular drive only. Dropping the second term is the *rotating-wave approximation*. Its accuracy depends on the ratio Ω/ν and on the time. For example, with $\Omega/\nu_0 = 0.01$ and $\nu = \nu_0$, a numerical solution of the Schrödinger equation shows that a spin prepared up is turned over with probability above 0.999 at the end of a pulse of duration π/Ω. The small shift of the resonance that the second term causes is computed on Exercise Sheet 2.

The driven qubit, as a quantum system

States and observables: those of Lecture 1, the points of the Bloch sphere and the self-adjoint matrices $a\mathbb{1} + b\,\mathbf{m}\cdot\boldsymbol{\sigma}$. *Evolution:* $\mathrm{i}\hbar\dot{\psi}_t = H(t)\psi_t$. Under the circular drive (2.1), $\psi_t = R(t)\, e^{-\mathrm{i}t\tilde{H}/\hbar}\psi_0$, and in the rotating frame the sphere turns about $(\Omega, 0, \Delta)/\tilde{\Omega}$

at the rate $\tilde{\Omega}$ (Lemma 2.1). *Measurement:* as in Lecture 1. Starting from $|\uparrow\rangle$, the reading $|\downarrow\rangle$ has probability $(\Omega/\tilde{\Omega})^2 \sin^2(\tilde{\Omega}t/2)$ (Theorem 2.2).

Summary.

- *The rotating frame*: the circular drive becomes a constant Hamiltonian, so for each initial vector the Schrödinger equation has exactly one solution, given in closed form (Lemma 2.1).
- *Rabi's formula*: under a circular drive with any $\Omega > 0$, a spin prepared up is turned over on resonance at the time π/Ω. Off resonance the probability of the turn stays below 1 (Theorem 2.2).
- *Ramsey's fringes*: from spin up, two ideal quarter turns about $\mathbf{e}_x$, a wait T apart with the drive's phase kept, turn the spin over with probability $\cos^2(\Delta T/2)$ (Proposition 2.4).

Lecture 3 asks what happens when the preparation is not known exactly, as for the atoms straight from the oven.

References

Cohen-Tannoudji, Diu and Laloë, *Quantum Mechanics*, vol. 1, Complement F_{IV}, pp. 455–461, and Ch. IV, §C-3-b, p. 419. Sakurai and Napolitano, *Modern Quantum Mechanics*, §§5.5.3–5.5.4, pp. 323–326. Weinberg, *Lectures on Quantum Mechanics*, §6.8, pp. 232–237. Rabi, Zacharias, Millman and Kusch, Phys. Rev. **53**, 318 (1938). Purcell, Torrey and Pound, Phys. Rev. **69**, 37 (1946). Bloch, Hansen and Packard, Phys. Rev. **69**, 127 (1946). Ramsey, Phys. Rev. **76**, 996 (1949), and Phys. Rev. **78**, 695 (1950). *Linear Algebra for Physics*, Prop. 6.21.